MILITARY SPECIFICATION

BATTERY, PRIMARY, LITHIUM-SULFUR DIOXIDE (FOR AN/PRC-96 RADIO TRANSCEIVER)

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and tests for production of the primary lithium-sulfur dioxide battery which provides power for the AN/PRC-96 search and rescue radio transceiver (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	Box, Folding, Paperboard

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-129	Marking For Shipment And Storage

PUBLICATIONS

MILITARY

NAVAL ELECTRONIC SYSTEMS COMMAND

NAVELEX 975	Electronics Information Bulletin

NAVAL SUPPLY SYSTEMS COMMAND

NAVSUP 4500	Consolidated Hazardous Item List

NAVY ENVIRONMENTAL SUPPORT OFFICE

NESO 20.2-011	Hazardous Waste Disposal Guide

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-5043), Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF TRANSPORTATION (DOT)

DOT-E-7052	Exemption From Hazardous Materials Regulation 49 CFR 172.101, 173.206(e)(1), 175.3

(Application for copies should be addressed to: Associate Director for Hazardous Materials Regulation, Materials Transportation Bureau, Research and Special Programs Administration, US Department of Transportation, Washington, DC 20590 Attention: Exemptions Branch.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM F134-78	Standard Recommended Practices For Determining Hermeticity Of Electron Devices With A Helium Mass Spectrometer Leak Detector

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3).

3.2 Construction. The battery shall consist of two, series connected, cylindrical lithium-sulfur dioxide cells placed end to end and assembled in a suitable paper, fiberboard, or plastic tube to form a right cylinder as shown in FIGURE 1.

3.2.1 Dimesions. The dimensions of the finished battery shall be within the limits shown on FIGURE 1. The finished battery shall pass through a straight cylinder 3.40 centimeters (cm) (1.340 inches (in)) maximum diameter and 12.70 cm (5 in) long without binding (see 4.6.1).

3.2.2 Weight. The finished battery shall weigh 170 ±28 grams (6.0 ±1 ounces) (see 4.6.2).

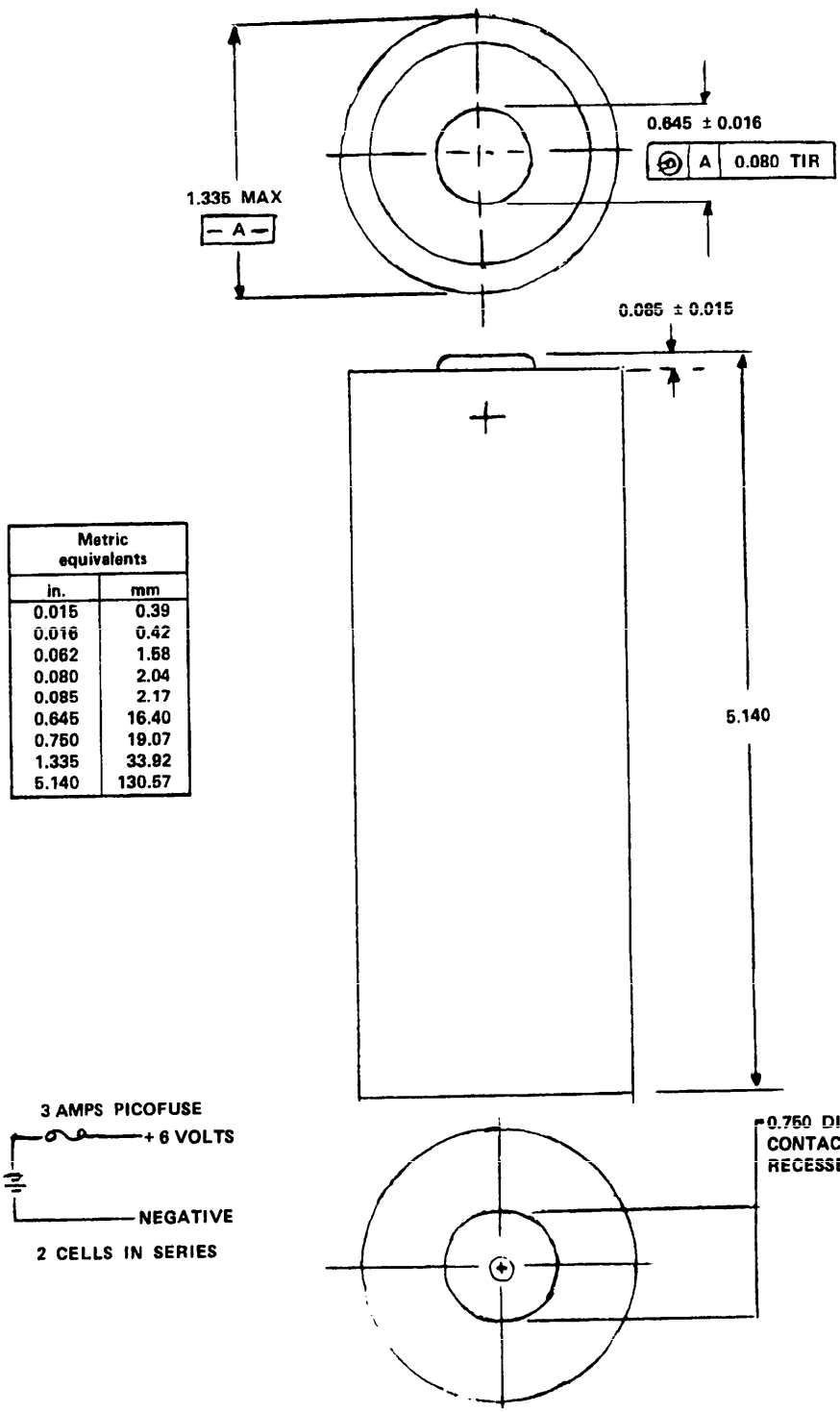
3.2.3 Safety. The safety devices and precautions for the battery shall conform in all respects to the guidelines specified in the Appendix to this specification and to the requirements contained herein. Any change in design or in the safety devices incorporated in the battery shall require a safety reevaluation in compliance with the Appendix.

3.2.3.1 Fuse. Each battery shall have a 3 ampere picofuse assembled as a part of the internal wiring of the battery.

3.2.3.2 Vent. Each cell of the battery shall have a vent that will open when the internal pressure of a cell reaches 300 to 500 pounds per square inch gage (psig). The battery shall be designed and constructed so that the vent is operable under all conditions of use and abuse (see 4.6.3.2).

3.2.3.3 Seals. Each cell of the battery shall be hermetically sealed. The seal between the metal cell cover and the metal can shall be a continuous weld and shall be free from holes and other imperfections. The seal between the center terminal and the cover shall be of the glass-to-metal or ceramic-to-metal type and shall be free from holes or imperfections. The hermeticity of the sealed cell shall be 5×10^{-8} atm.cm^3/second maximum (see 4.6.3.3).

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Metric equivalents	
in.	mm
0.015	0.39
0.016	0.42
0.062	1.58
0.080	2.04
0.085	2.17
0.645	16.40
0.750	19.07
1.335	33.92
5.140	130.57

NOTES:

1. Dimensions are in inches.
2. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based on 1 inch = 25.4 mm.

FIGURE 1. Battery.

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3.2.3.4 Marking. The labeling or marking on the battery shall conform to that shown in FIGURE 2. The marking shall be clearly legible throughout the life of the battery (see 4.6.3.4).

3.2.3.5 Storage. When possible, batteries shall be stored in the upright position. The battery after storage shall be capable of fulfilling the requirements of TABLE I (see 4.7.6.1, 4.7.6.2, and 6.3.1).

3.2.3.6 Disposal. Disposal instructions shall be included with each battery and shall conform to the requirements of FIGURE 2 and 5.1.1.1.1.

3.3 Electrical. The battery shall conform to all of the electrical requirements specified herein.

3.3.1 Open circuit voltage. The open circuit voltage of the battery shall be within the range of 5.83 to 6.10 volts at 20° Celsius (C) (68° Fahrenheit (F)). Testing shall be in accordance with 4.6.4.1.

3.3.2 Rise time. When measured within the range of -10°C (14°F) and 71°C (160°F) and at any time from the date of manufacture to five years, the voltage of the battery shall rise to a minimum of 4.8 volts within five seconds after the start of the discharge through a 20 ±0.5 ohm circuit. Testing shall be in accordance with 4.6.4.2.

3.3.3 Discharge capacity. The discharge capacity of the battery to an end point of 4.0 volts, shall conform to the requirements in TABLE I. Testing shall be in accordance with 4.6.4.3.1, 4.6.4.3.2, 4.7.6.1, and 4.7.6.2.

TABLE I. Discharge and storage requirements.

Storage conditions		Discharge conditions		
Time (days)	Temperature °C	Temperature °C	Current mA	Minimum run time (hours)
0 to 365	-54 to 54	5 to 71 -10 to 5	350 400	20 16
366 to 1830 1/	-54 to 54	5 to 71 -10 to 5	350 400	16 13

1/ See 6.3.1

3.3.4 Electrolyte leakage. The battery shall show no sign of electrolyte leakage at any time during the life of the battery (see 4.6.5).

3.4 Environmental requirements. The batteries shall conform to the requirements of TABLE I after being subjected to the environmental conditioning requirements listed herein, in TABLE II, as described in DOT-E-7052, and as modified herein (see 4.7).

a. Altitude
b. Thermal stability
c. Vibration
d. Shock
e. Random drop
f. Storage life and discharge (365 days and 730 days)

3.5 Workmanship. The battery shall be constructed and finished in a manner to ensure compliance with all the requirements of this specification. Particular attention shall be paid to the seals, the electrochemical ratio of active materials, the assembly of the electrodes and separator, the control of moisture, and the reliability of the fuses. All welds, solder joints, and fill-tubes and seals shall be free from any defect that could adversely affect the performance of the battery. Heat shrinkable plastic tubing, when used, shall be applied in a manner to preclude wrinkles or other defects (see 3.2.1).

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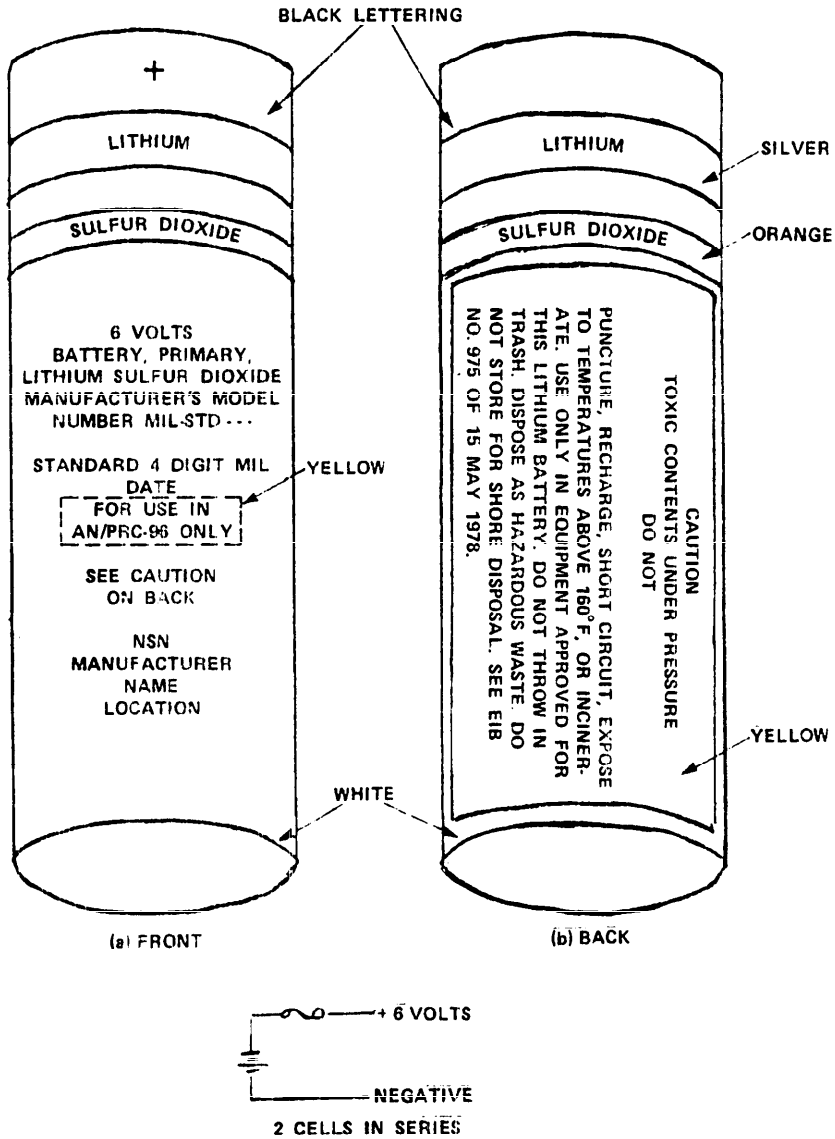


FIGURE 2. Battery labeling.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. Unless otherwise specified (see 6.2), 12 batteries, complete in every detail and manufactured by the equipment and processes that will be used to manufacture the production lots of batteries, shall be submitted to a test facility, that will be designated in the contract, for first article inspection specified in TABLE II. Acceptance of the first article sample qualifies the design, materials, fabrication procedures and the specific facilities by which manufacture and test of the sample is accomplished. After manufacture and acceptance of the first article sample, no changes shall be made in the design, materials, or fabrication procedures without the approval of the procuring activity. When new or altered facilities are used in production, a new first article sample shall be submitted and inspected for acceptance as specified herein. Failure of one component or of any one battery on any test, TABLE II Tests 1 through 19 (inclusive), shall result in the rejection of the entire sample. Tests 4 through 18, 20, 21, 23, and 24 are non-destructive. The purpose of tests 20 through 25 (inclusive) is to provide timely evidence that the manufacturer can fulfill all of the requirements of TABLE I (see 6.3). Release to production shall be granted on the acceptable performance on tests 1 through 19 (inclusive) TABLE II.

4.4 Quality conformance inspection. Unless otherwise specified herein, and when applicable, the sampling plans and procedures used in determining the acceptability of products shall be in accordance with MIL-STD-105 (see 4.4.1). Inspection of each lot of batteries submitted for acceptance shall be performed by the contractor and an activity designated by the procuring activity. The inspection shall consist of the examination and tests specified in TABLE III.

4.4.1 Sampling for quality conformance inspection. At the discretion of the Government, a sample of batteries may be selected from each production lot (see 4.4) for determining conformance to the storage and discharge requirements of TABLE I, 4.7.6.1, and 4.7.6.2 to determine if the contractor can continue to produce batteries that can fulfill the storage requirements of TABLE I.

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TABEL II. First article inspection.

Inspection or test	Sample size	Requirement paragraph	Test method paragraph	Comments
1. Fuse 1/ 5/	12	3.2.3.1	4.6.3.1	
2. Vent 1/ 5/	12	3.2.3.2	4.6.3.2	
3. Seals 1/ 5/	12	3.2.3.3	4.6.3.3	
4. Electrolyte leakage and workmanship	12	3.3.4 and 3.5	4.6.5	
5. Marking	12	3.2.3.4	4.6.3.4	
6. Dimensions	12	3.2.1	4.6.1	
7. Weight	12	3.2.2	4.6.2	
8. Open circuit voltage	12	3.3.1	4.6.4.1	Tests 8 through 14 must be run consecutively
9. Rise time 2/	12	3.3.2	4.6.4.2	
10. Altitude	12	3.4	4.7.1	
11. Thermal stability	12	3.4	4.7.2	
12. Vibration	12	3.4	4.7.3	
13. Shock	12	3.4	4.7.4	
14. Random drop	4	3.4	4.7.5	Two units assigned to Test 19 and one each to 22 and 25
15. Electrolyte leakage and workmanship	12	3.3.4 and 3.5	4.6.5	
16. Dimensions	12	3.2.1	4.6.1	
17. Open circuit voltage	12	3.3.1	4.6.4.1	
18. Rise time	12	3.3.2	4.6.4.2	Immediately preceding Test 19
19. Discharge 1/ 3/ (No storage)	6	3.3.3	4.6.4.3.1 and 4.6.4.3.2	Two units from Test 14 one on 4.6.4.3.1 and one on 4.6.4.3.2
20. Open circuit voltage 4/	3	3.3.1	4.7.6.1 and 4.6.4.1	Immediately preceding Test 21
21. Rise time 4/	3	3.3.2	4.7.6.1 and 4.6.4.2	Immediately preceding Test 22
22. Discharge (after 365 days storage) 1/ 4/	3	3.4	4.7.6.1	One unit from random drop Test 14

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TABLE II. First article inspection. (continued)

Inspection or test	Sample size	Requirement paragraph	Test method paragraph	Comments
23. Open circuit voltage 4/	3	3.3.1	4.6.4.1	Immediately preceding Test 24
24. Rise time 4/	3	3.3.2	4.7.6.1 and 4.6.4.2	Immediately preceding Test 25
25. Discharge (after 730 days storage) 1/ 4/	3	3.4	4.7.6.2	One unit from random drop, test 14

1/ Test is destructive.

2/ Six units shall be tested at -10°C ±1°C. The remaining units shall be tested at 71°C ±1°C.

3/ One of the units from test 14 shall be discharged in accordance with 4.6.4.3.2 at -10°C +1°C. The remaining units from test 14 shall be discharged in accordance with 4.6.4.3.1 at 5°C ±1°C. The remaining four units shall be discharged as follows:

One unit at 5°C in accordance with 4.6.4.3.1
One unit at 71°C in accordance with 4.6.4.3.1
One unit at -10°C in accordance with 4.6.4.3.2
One unit at 5°C in accordance with 4.6.4.3.2

4/ See 4.3 and 6.3

5/ To be performed at the contractor's facility on a sample of components prior to the fabrication of the first article sample.

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TABLE III. Quality conformance inspection performed by an activity designated by the procuring activity.

Inspection or test	Requirement paragraph	Test method paragraph	Number of units tested or MIL-STD-105 inspection level	Number of failures or AQL percent defectives
1. Electrolyte leakage and workmanship	3.3.4 and 3.5	4.6.5	100 percent	Reject all defectives
2. Marking	3.2.3.4	4.6.3.4	II	1 percent
3. Dimensions	3.2.1	4.6.1	100 percent	Reject all defectives
4. Weight	3.2.2	4.6.2	II	1 percent
5. Open circuit voltage	3.3.1	4.6.4.1	100 percent	Reject all defectives
6. Rise time	3.3.2	4.6.4.2	II	1 percent
7. Altitude	3.4	4.7.1	2/	
8. Thermal stability	3.4	4.7.2	2/	
9. Vibration	3.4	4.7.3	2/	
10. Shock	3.4	4.7.4	2/	
11. Random drop	3.4	4.7.5	2/	
12. Open circuit voltage	3.3.1	4.6.4.1	S-2	1 percent
13. Rise time	3.3.2	4.6.4.2	S-2	1 percent
14. Discharge 1/ (No storage)	3.3.3	4.6.4.3.1 and 4.6.4.3.2	S-2	1 percent
15. Fuse 1/ 3/	3.2.3.1	4.6.3.1	2/	
16. Vent 1/ 3/	3.2.3.2	4.6.3.2	2/	
17. Seals 1/ 3/	3.2.3.3	4.6.3.3	2/	

1/ Test is destructive. Do not return test units to lot.

2/ Number of units for environmental conditioning shall be based on sampling in accordance with MIL-STD-105; Inspection Level S-2, but no accept or reject criteria shall apply.

3/ To be performed at the contractor's facility prior to production run.

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4.5 Test equipment, facilities, and conditions.

4.5.1 Test equipment and facilities. The contractor shall furnish and maintain all necessary test equipment and facilities for performing the first article and quality conformance inspection of TABLES II and III. The test equipment shall be adequate in quantity, and when definite requirements are not specified, the equipment shall be of sufficient accuracy and quality to permit performance of the required tests.

4.5.2 Test conditions. Unless otherwise specified herein, all inspections shall be performed at room ambient conditions.

4.6 Test procedures.

4.6.1 Dimensions. The dimensions of the completed battery shall conform to the requirements of 3.2.1. Particular attention shall be given to the requirement for passing through the straight cylinder without binding. Care shall be exercised to avoid shorting the battery if metal measuring devices are used.

4.6.2 Weight. The battery shall be weighed to the nearest ounce to verify conformance to the requirements of 3.2.2.

4.6.3 Safety.

4.6.3.1 Fuse. Each fuse in the sample shall be electrically connected to a six-volt DC power source. The current shall be gradually increased to five amperes. Any fuse that opens at less than three amperes or fails to open within one hour when the current reaches four amperes or within five seconds when the current reaches five amperes shall be rejected and the first article sample or production lot shall be rejected (see 3.2.3.1).

4.6.3.2 Vent. Each cell case in the sample shall be tested for venting pressure. The samples shall be tested under a hydraulic oil pressure using the typical test set-up shown in FIGURE 3. The pressure shall be gradually increased until the vent opens. The vent shall remain closed at 300 psig but shall open at a pressure no greater than 500 psig. Failure of the vent to open within the limits specified herein shall cause rejection of the first article sample or production lot (see 3.2.3.2).

4.6.3.3 Seals. The can and cover shall be assembled and sealed without electrodes, separators, and electrolyte. The sample units shall be filled with helium gas under a slight positive pressure and the fill tube welded. The sample units shall then be tested by a mass spectrometer as described in ASTM F134-78. The hermeticity shall be reported together with the location of any leak that may be detected. Failure to conform to the requirements of 3.2.3.3 shall result in rejection of the first article sample or production lot.

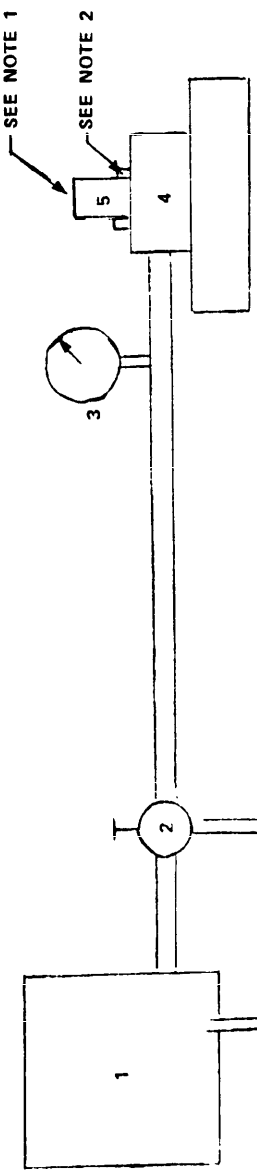
4.6.3.4 Marking. The marking shall conform to the requirements of 3.2.3.4 as shown in FIGURE 2.

4.6.4 Electrical.

4.6.4.1 Open circuit voltage. The open circuit voltage shall be measured on a voltmeter having a resistance of 20,000 ohms per volts and an accuracy of ±2 percent of full scale. The voltage shall conform to the requirements of 3.3.1.

4.6.4.2 Rise time. The rise time shall be determined using a circuit with a resistance of 20 ±0.5 ohms. Five seconds after closing the switch, the voltage across the battery shall be a minimum of 4.8 volts. The tests shall be terminated after the requisite five seconds have elapsed (see 3.3.2).

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EQUIPMENT:

1. Hydraulic pump
2. By-pass valve—to control and regulate the flow of oil to the test sample.
3. Pressure gage—to permit visual indication of pressure to the test sample.
4. Sample holding fixture.
5. Sample cell case.

NOTES:

1. Cell case shall be turned bottom side up with topside of cell case open to the oil pressure.
2. Pressure and leaktight seal area.

FIGURE 3. Vent test set-up, typical.

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4.6.4.3 Discharge capacity.

4.6.4.3.1 Discharge 5° Celsius (C) to 71° C (no storage). The battery shall be discharged at a constant current of 350 milliamperes (mA) ±2 percent to an end-point of 4.0 volts. The battery shall be conditioned at the test temperature for a minimum of 10 hours prior to the start of the discharge. The temperature surrounding the battery shall be maintained at the conditioning temperature during discharge (see 3.3.3).

4.6.4.3.2 Discharge, -10° C to 5° C, (no storage). The battery shall be discharged at 400 mA ±2 percent to an end point of 4.0 volts. The battery shall be conditioned at the test temperature for a minimum of 10 hours prior to the start of the discharge. The temperature surrounding the battery shall be maintained at the conditioning temperature during discharge (see 3.3.3).

4.6.5 Electrolyte leakage and workmanship. Each finished battery in the sample shall be visually examined for any indication of fluids that might indicate electrolyte leakage. In addition, the visual examination shall be employed to indicate conformance with the workmanship requirements of 3.5 (see 3.3.4).

4.7 Environmental conditioning. The battery shall be subjected to the environmental conditons listed in 3.4, in the sequence specified in 4.7 through 4.7.6.2 and as shown in TABLES II and III.

4.7.1 Altitude. The sample shall be subjected to the altitude conditioning described in DOT-E-7052.

4.7.2 Thermal stability. The sample shall be subjected to the thermal stability conditioning described in DOT-E-7052.

4.7.3 Vibration. The sample shall be subjected to the vibration conditioning described in DOT-E-7052.

4.7.4 Shock. The sample shall be subjected to the shock conditioning described in DOT-E-7052.

4.7.5 Random drop. The battery shall be mounted in a case simulating the AN/PRC-96 radio and weighing 1,360.7 grams (48 ounces) ±56.7 grams (2 ounces). The assembly shall be subjected to 10 random drops from a height of 182.88 cm (6 feet(ft)) ±15.24 cm (0.5 ft) onto a 1.27 cm (0.50 in) thick steel plate.

4.7.6 Storage life.

4.7.6.1 Discharge 5°C to 71°C (after 365 days storage). After the non-destructive tests and the environmental conditioning listed in TABLE II plus 365 days of storage within the temperature range of -54°C to 54°C, the open circuit voltage (4.6.4.1) and the rise time (4.6.4.2) shall be determined on the batteries selected for this test. The batteries shall then be conditioned and discharged in accordance with 4.6.4.3.1 (see 4.4.1 and 6.3.1).

4.7.6.2 Discharge -10°C to 5°C (after 730 days storage). After the non-destructive tests and the environmental conditioning listed in TABLE II plus 730 days of storage within the temperature range of -54°C to 54°C, the open circuit voltage (4.6.4.1) and the rise time (4.6.4.2) shall be determined on the batteries selected for this test. The batteries shall then be conditioned and discharged in accordance with 4.6.4.3.2 (see 4.4.1 and 6.3.1).

4.8 Visual examination and failure criteria. Each battery shall be visually examined and a record made of any damage or deterioration resulting from each test. Any evidence of electrolyte leakage, outgassing, deterioration, or change in physical tolerance limits which could in any manner prevent the battery from conforming to operational service requirements shall constitute a failure and shall be a cause for rejection.

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5. PACKAGING

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with DOT-E-7052 and 5.1.1 through 5.2.2.

5.1.1 Preservation and packing.

5.1.1.1 Inner packaging. Each battery shall be enclosed in a heat sealed plastic bag (polyethylene - 4 mil thick) and packed in an inner paperboard box that complies with PPP-B-566.

5.1.1.1.1 A sheet with the following information shall be packed with each battery in the inner box: In accordance with Electronics Information Bulletin No. 975, mark each used battery discharged: Do Not Use. Individually place in a plastic bag and seal or individually wrap in electrical insulating material. For disposal at sea, discharged batteries should be removed from the plastic bag and dropped overboard in deep water ((in excess of 152.4 meters (500 ft)) outside the prohibited zone ((80 kilometers (50 miles) limit)). Do not store for shore disposal. For shore disposal lithium batteries shall be buried in controlled hazardous waste landfills (see NESO 20.2-011).

5.1.1.2 Outer packaging. One battery packaged as specified in 5.1.1.1 shall be packed in an outer paperboard box that complies with PPP-B-566.

5.2 Marking.

5.2.1 Special markings. Both inner and outer boxes, and shipping container shall be properly marked indicating contents, quantity, Store This Side Up (indicating top of battery), manufacturer's part number, and identification number. The words CAUTION, CONTAINS LITHIUM BATTERIES -- DO NOT EXPOSE TO TEMPERATURE ABOVE 71.1°C (160°F) HANDLE WITH CARE shall be included on both inner and outer boxes markings. In addition, the inner and outer boxes and shipping container shall be marked with the Hazardous Material Marking Symbol of NAVSUP Publication 4500, Fire Hazard-1, Reactivity-1, Specific Hazard - Cor, Health Hazard-2.

5.2.2 Normal markings. In addition to any special markings required by the contract, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129 and DOT-E-7052.

6. NOTES

6.1 Intended use. The intended use for the battery covered by this specification is to power the AN/PRC-96 radio onboard ships.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Type and degree of contractor quality assurance system required (see MIL-Q-9858)
c. Whether or not the test samples required are included in the number of batteries ordered
d. Number of first article samples to be submitted if other than specified in 4.3
e. Level of preservation, packaging, packing, and marking required (see Section 5)
f. Special marking requirements (see 5.2.1)
g. Instructions for conducting storage life tests (see TABLE I, 4.7.6.1, and 4.7.6.2) as part of first article tests (see TABLE II), if required (see 4.3 and 6.3)

6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation (DAR). The first article should be a first production item. The first article should consist of 12 units. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

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6.3.1 Instructions for first article. Actual one-year (365 days), two-year (730 days) and five-year (1830 days) storage tests are not required by this specification for acceptance of the first article sample (see 4.3). If required by the contract, samples from the first article tests for TABLE II, Tests 20 through 25, shall be set aside for determining conformance to the storage and discharge requirements of TABLE I, 4.7.6.1, and 4.7.6.2. The Government assumes that successful completion of the two-year storage tests of TABLE II, Test 25, is a strong indicator that the five-year storage requiremnts of TABLE I can be achieved. However, the Government reserves the right to perform the five-year storage tests to determine the validity of the battery storage requirements (see 3.2.3.5).

Preparing activity:
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APPENDIX

GUIDELINES FOR DESIGN, ACQUISITION,
USE, PACKAGING, TRANSPORT, AND DISPOSAL OF LITHIUM BATTERIES

10. SCOPE

10.1 Scope. This appendix provides the guidelines for the design, acquisition, use, packaging, transportation, and disposal of lithium batteries.

20. APPLICABLE DOCUMENTS

20.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

DoD-D-1000	Drawings, Engineering And Associated Lists

STANDARDS

MILITARY

MIL-STD-480	Configuration Control - Engineering Changes, Deviations And Waivers
MIL-STD-648	Design Criteria For Specialized Shipping Containers

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. REQUIREMENTS

30.1 Only lithium batteries which have been approved as safe shall be acquired for use. The Systems Command having cognizance of the development or acquisition shall be responsible for issuing such approval. A technical evaluation of the battery and its intended use shall be the basis for approval decisions. The Naval Surface Weapons Center (NAVSURFWPNCEN), White Oak, MD (Electrochemistry Branch) should be utilized for this technical evaluation. Programs anticipating the use of lithium batteries shall submit to the NAVSURFWPNCEN, via the Commander, Naval Sea Systems Command, (COMNAVSEASYSCOM), Washington, DC 20362, a data package validating the selection of the lithium battery by describing: a) the proposed battery, for example, design, geometry, and electrochemical system, b) the equipment design, current drain, types of safety features, fuses (include valves), battery use, venting and strength, c) the logistic and operational use sequency of the item in which the battery is to be used. Upon completion of the technical review, a recommendation concerning the safety of the proposed battery will be presented to the cognizant command by COMNAVSEASYSCOM. In cases warranting such action, the Weapon System Explosives Safety Review Board shall be convened to review the technical evaluation and make recommendations concerning the safety of the item in which the battery is to be used.

30.2 All lithium batteries shall be marked to indicate the information required by FIGURE 2. In addition to this information, the unit packages shall be marked with the Hazardous Material Marking Symbol of NAVSUP Publication 4500.

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30.3 Manufacturers shall be required to provide Material Safety Data Sheets in accordance with DAR 7-104.98.

30.4 All unit cells shall be constructed so that the cell case to cover seal is a continuous weld, free from holes and other imperfections. The seal between the positive electrode and the cover shall be of the glass or ceramic-to-metal type and free from imperfections. Each cell, battery, and battery enclosure must incorporate a safety venting device or be designed and manufactured in such a manner that will preclude a violent rupture condition, such as dead short or involvement in a fire, incident to transportation, use, storage or disposal. Nothing shall be done in the design and construction of the battery that will degrade the vent.

30.5 Each cell used as the power source, or each group of cells connected in series, shall contain a fuse or other suitable over-current device that will open if that group of cells is discharged at an excessive rate. When possible, heat and pressure sensing devices shall be employed which activate an electrical bypass of a defective cell.

30.6 To prevent shorts, internally and externally, all metal parts shall be secured to prevent any movement at any time during a battery lifetime. Electrode separators shall be compatible with the other materials of construction.

30.7 Whenever possible, lithium batteries shall be constructed with cells of a size that are not interchangeable with commercial flashlight and radio batteries.

30.8 Whenever possible in development programs, completely assembled batteries should be procured from a battery manufacturer. Assembly of batteries by inexperienced personnel should be avoided.

30.9 Basic packaging, marking, and shipping requirements imposed by the Department of Transportation are contained in DOT-E-7052. In addition to the minimum requirements of DOT-E-7052, activities desiring to use lithium batteries shall:

a. Ensure that a complete design disclosure is obtained on the packaging of the specific battery, preferably prior to any shipment, but in any case, no later than release for limited production or full scale production whichever occurs first.
b. Ensure that the design disclosure is incorporated in the appropriate acquisition specification, contract, and manuals. Descriptive specification language shall be supplemented by DoD-D-1000 drawings or figures in the specification, as appropriate.
c. Ensure that the efficiency of the packaging is demonstrated by tests and obtain a test report. The minimum package performance level is contained in MIL-STD-648. Other tests required by DOT-E-7052 shall also be performed.
d. Ensure that where batteries are entered in the supply system for organizational or intermediate maintenance level replacement, batteries so acquired are packaged so as to be capable of shipment by cargo only aircraft.
e. Advise potential contractors not listed in the latest issue of DOT-E-7052 that they must become a party thereto prior to shipping batteries by any mode.

30.10 Activities procuring batteries for limited or full scale production shall ensure that configuration management is imposed on the battery and its packaging by use of MIL-STD-480. In addition to the usual definition, a Class 1 change shall be defined as any change affecting safety characteristics of the battery as described herein. Class 1 battery changes shall be coordinated with NAVSURFWPNCEN. Class 1 packaging changes shall be reviewed by personnel formally qualified in hazardous material packaging.

30.11 Storage of lithium batteries shall be as follows:

a. Lithium cells and batteries shall be stored in their original shipping containers in a cool, ventilated shelter.
b. The storage area shall be isolated from other hazardous and combustible material and used only for the storage of unused lithium batteries.

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c. Since the effect of mass storage on the hazard degree is not known, the quantity stored in an area shall be kept to the minimum consistent with requirements.
d. Batteries in storage shall be retained in unit packages, preferably shipping containers, to prevent heat transfer between batteries.

30.12 Rough handling of lithium batteries may increase the hazards in use. Therefore, qualification testing for a specific application should include environmental testing representative of the actual environments to be encountered in the logistic cycle for that application.

30.13 In the event of an accident, incident or malfunction, either with or without visible damage to the battery, notify the appropriate authorities.

30.14 Since most of the documented safety incidents have occurred with a partially or fully discharged lithium battery, such should be removed from its associated equipment upon completion of its useful life. The exposed terminal shall be insulated to prevent short circuits, and the battery shall be turned in to an appropriate disposal agency. When possible, agencies licensed to dispose of hazardous material shall be utilized. If this is not possible, then dispose as follows:

a. At sea, discharge overboard in deep water (in excess of 152.4 meters (500 ft)) outside the prohibited zone (80.45-kilometers (50-mile limit)). Do not store for shore disposal.
b. Ashore, batteries shall be buried in a controlled hazardous waste landfill. Disposal directions may be obtained from the Navy Environmental Support Office (see NESO 20.2-011 and the DOD Disposal Guidelines).

30.15 Used lithium batteries shall not be allowed to accumulate and disposal shall be effected promptly. A collection point and storage area separate from other combustible material shall be established for batteries awaiting disposal. Lithium batteries shall not be disposed of nor transported with normally generated refuse.

30.16 Whether fresh or discharged, lithium batteries shall not be pierced, crushed, burned, intentionally dropped, cannibalized, dismantled, modified, or otherwise carelessly handled, nor shall they be short circuited, charged, or used in any way other than their intended use.

30.17 An extinguisher suitable for metal fires (class D fires), for example, MET-L-X or LITH-X, shall be used to extinguish burning lithium. When such a dry powder type fire extinguisher is used, some of the extinguishing agent may spread throughout a wide area around the point of use. Therefore, if at all possible, remove items such as electronic instruments which could be contaminated by the powder particles (sodium chloride base with MET-L-X and graphite base with LITH-X) from any area where a dry powder type extinguisher might be used. This guidance is primarily for battery assembly facilities, since the design and size of currently contemplated lithium batteries for fleet use will preclude the occurrence of lithium metal fires.

30.18 When entering a storage space in which lithium batteries have vented gases, supplied air respirators or self-contained breathing apparatus approved by the National Institute for Operational Safety and Health (NIOSH) shall be worn.

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MATERIAL PROCURED UNDER A
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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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